

Unit 9, Lesson 1: Adding and Subtracting Linear Expressions with Integer Coefficients

Benchmark

MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

Learning Target

- ☐ I can use algebra tiles to add and subtract linear expressions.

9.1.1 Warm-Up: Career in Robotics

Tessa Lau is the Chief Robot Whisperer for a robotics company. The company creates robot helpers for the service industry. Some of their robots perform tasks like delivering items to guests at a hotel. Tessa has studied applied physics, computer science, artificial intelligence, and human-computer interaction. She loves hands-on experiments, building things, and the speed of creating with computers. Through programming and engineering, one of her primary jobs is to look at a robot in context and make sure the robots behave well in human environments.



1. Have you ever considered a career as an engineer?
2. How do you think a robot could be helpful to you in your everyday life?
3. Tessa's advice is "Don't listen to someone who tells you that you can't do something." Describe a time someone told you that you were not able to do something.

9.1.2 Refresher: Adding and Subtracting Integers with Algebra Tiles

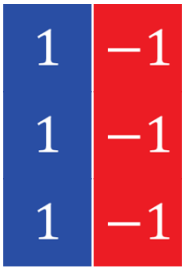
Algebra tiles, like integer chips, can be used to model adding and subtracting integers. Each small square represents a positive 1 (blue) or a negative 1 (red).

Integer Chips	Algebra Tiles
 	 

Adding a positive 1 (blue) and a negative 1 (red) represents a zero pair because $1 + ^{-}1 = 0$, as shown.



1. Describe why this diagram represents the value 0.



A **zero pair** is a pair of numbers whose sum is zero.

2. Complete the following problems.

- a. Complete the table.

Numeric Expression	Algebra Tiles Representation	Value
$-7 + 2$		
$6 - 8$		
$-1 - 5$		
$4 - (^{-}3)$		

- b. For some of the representations with algebra tiles, zero pairs had to be added to the representation. Put an asterisk (*) next to the expressions where this was the case.
- c. Explain why zero pairs had to be added to the representation in these cases.

9.1.3 Guided Instruction: Algebra Vocabulary

There are some fundamental definitions that are necessary to know for working with algebraic expressions.

A **term** is a single number or variable, or numbers and variables multiplied together. For example, the expression $0.5 - 4y$ has two terms.

The expression $0.5 - 4y$ has two terms.

0.5	−4y
term 1	term 2

One of the terms in this expression is a constant term.

1. Which term is the constant term? Explain.



An **expression** is a mathematical statement containing numerals, operators, grouping symbols, and symbols or variables for unknown values. An expression does not contain an equal sign or inequality symbol.

For example, $4 + 5$, $3x - 2$, and $6x - 7y + 11$ are all expressions.



A **coefficient** is the number or constant that multiplies a variable in an algebraic expression. If no number is specified, the coefficient is 1.

In the expression $4xy$, 4 is the coefficient.

In the expression x , 1 is the coefficient.

In the expression $-8y$, -8 is the coefficient.

Variable terms have coefficients while constant terms do not.

- d. Circle the zero pairs in your drawing.
- e. Complete the expression below after eliminating the zero pairs.

$$\underline{\hspace{1cm}}x + \underline{\hspace{1cm}}y + \underline{\hspace{1cm}}$$

Like terms are terms with the exact same variable(s) raised to the same exponent.

A **constant** is a fixed numerical value with no variables. All constants are like terms.

2. Answer the following questions about like terms.

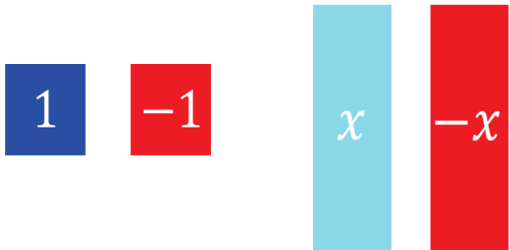
- a. Classify the following terms.

	Like Terms	Unlike Terms
y and 6	☐	☐
$3x$ and 3	☐	☐
$5x$ and $-3x$	☐	☐
10, -2 , 7.5	☐	☐

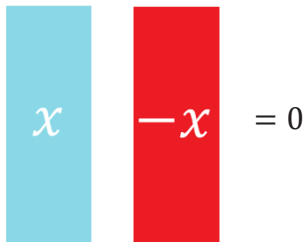
- b. How did you determine whether the terms were like or unlike?
- c. Compare answers with your partner.

9.1.5 Guided Instruction: Adding and Subtracting Expressions with Algebra Tiles

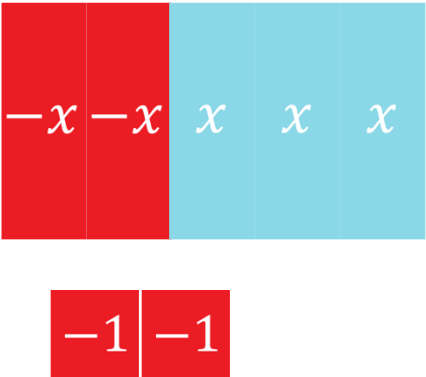
The algebra tiles shown represent x and $-x$ as rectangles, and integers as squares. Red represents the negative of that variable or constant, while the blue shades represent the positive.



Variables also form zero pairs, as shown.



1. Complete the table by crossing out the zero pairs and recording how many there are. Write the resulting algebraic expression after eliminating the zero pairs.

Algebra Tiles	Number of Zero Pairs	Algebraic Expression
Representation: 		

2. Draw a representation of the following expressions with algebra tiles. Cross out any zero pairs. Then write an equivalent algebraic expression after eliminating any zero pairs.

Algebraic Expression and Representation	Equivalent Algebraic Expression
$x - 3x + 1$ Representation:	
$-4 + 3y - y + 2$ Representation:	

9.1.6 Guided Practice: Adding and Subtracting Expressions with Algebra Tiles

1. Consider the expressions $2x$ and $x + 2$.
- a. Draw a model of $2x$ and $x + 2$ with algebra tiles.

Expression	Algebra Tiles Model
$2x$	
$x + 2$	

- b. How are they different?
- c. Are $2x$ and $x + 2$ equivalent algebraic expressions?

9.1.7 Your Turn!

Write an equivalent expression by combining like terms.

1. $2x + 4x =$ _____

2. $-y + 7 + 4y - 1 =$ _____

3. $x - x + x - x + 1 =$ _____

4. $8x - 4 - 4x =$ _____

5. $2x - 4x =$ _____

9.1.8 Wrap-Up: Cool Down

- For each key word, check the box in the column that describes your knowledge of the word. Then answer the questions or complete the statements in the last column.

Key Word	I already knew the meaning of this word.	I've heard this word before, but I was not sure what it meant.	Questions
Expression	☐	☐	An example of a numeric expression is _____. An example of an algebraic expression is _____.
Term	☐	☐	In the expression $8x - 9 + 2y$, how many terms are there? _____
Coefficient	☐	☐	In the expression $8x - 9 + 2y$, what are the coefficients? _____
Equivalent	☐	☐	When working with algebraic expressions the word "equivalent" means _____ _____ _____.

